Faycal Kilali

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Education

Université de Montréal – M.Sc Computer Science (with Internship) – Montreal, QC 2026 – 2028

- Graduate program with coursework drawn from **MILA (Quebec AI Institute)**-affiliated offerings taught by UdeM and McGill faculty, spanning deep learning, representation learning, self-supervised and meta-learning, reinforcement learning, generative models (GANs, GFlowNets, diffusion models), natural language processing and large language models, computer vision, probabilistic inference, optimization, graph and geometric deep learning, and AI applications in science and robotics.

Mount Allison University – B.Sc. (Hons) Computer Science and Mathematics – Sackville, NB 2022 – 2026

- **Certifications:** *Machine Learning with Python* (IBM), *Object Oriented Design* (UAlberta), *Microsoft Learn AI Skills* (Microsoft Corporation)
- **Founder and President** of the AI & Robotics Society – Sackville, NB 2023-2026
- **Co-Founder and Vice President** of the Mature Students Club – Sackville, NB 2023-2025

Experience

Fullstack Software Developer, AI Object Detection Framework and Educational Tool – Sackville, NB May 2024 – Sep 2024

- Secured $4,000 TD Ignite Grant and led development of AI-based object detection educational tool, implementing YOLOv10 model across Flask backend, Streamlit web interface, and Tkinter GUI with support for 243 languages
- Architected full-stack educational game integrating real-time computer vision, REST API design, and multi-platform deployment, achieving comprehensive documentation with API specifications and workflow diagrams

Mathematics and Computer Science Peer Tutor, Mount Allison University – Sackville, NB Mar 2023 – Sep 2024

- Tutored 50+ students across various mathematical subjects effectively, resulting in positive feedback from students

Technical Lead, Greater Dorchester Moving Forward – Dorchester, NB Jan 2023 – August 2023

- Spearheaded technology operations for a nonprofit organization by leading data collection, mapping, and software development–enhancing impact and advancing the mission
- Collaborated with other departments within the organization to integrate technical solutions relevant to the seated board

Selected Projects

Content-Aware Enterprise Document Search & Management System | Java, PostgreSQL (SQL), JDBC, PDFBox, Bash, MVC, Agile, Scrum, DevOps, Figma | GitHub 2025

- **Architected** a decoupled **MVC** system using **Dependency Injection** to orchestrate state management between the front-end and the application's backend, significantly reducing component coupling and enhancing system modularity
- **Designed** a hybrid search algorithm combining PostgreSQL's `tsvector` for optimized metadata retrieval ($O(\log n)$) with a custom Java-based regex engine for deep content analysis of binary PDF blobs, enabling granular pattern matching
- **Orchestrated** the team's **Agile/Scrum** workflow, serving as **Technical Lead** to mentor peers on Git best practices, resolve merge conflicts, and manage technical debt to ensure on-time delivery of prototypes
- **Established** comprehensive **DevOps** and documentation standards, authoring UML architectural diagrams and configuring **CI/CD** pipelines (Gradle) to standardize the build and release lifecycle for the team

Room Scanner & Voice Guidance System | Python, YOLOv10, Raspberry Pi, Bash, Computer Vision, Soldering, Embedded Engineering| GitHub 2024

- Engineered embedded room scanning system integrating computer vision with custom hardware circuits (GPIO, LED indicators, push-button controls), implementing both hardware and software debouncing techniques
- Developed real-time voice feedback system for object detection with respect to the physical location of the system, providing information/guidance for blind individuals throughout the observable area
- Created comprehensive documentation of the hardware schematics and the software components with automated installation scripts, thereby emphasizing reproducible deployment

Multi-Paradigm Wordle Puzzle Solver AI Agent | Python, Streamlit, Information Theory, AI | GitHub 2024

- Implemented four AI paradigms (Information-Theoretic, Model-Based, Goal-Based, Utility-Based), utilizing entropy calculations and information gain maximization to achieve optimal 3-5 guess performance
- Architected Streamlit visualization dashboard displaying real-time AI decision-making with time complexity of $O(W^2L)$, demonstrating advanced algorithm optimization and probabilistic reasoning

Multivariate Regression Framework | Python, PyTorch, NumPy, Machine Learning, AI | GitHub 2024

- Developed machine learning framework from mathematical foundations, implementing gradient descent optimization, L2 regularization (Ridge), and closed-form solutions using PyTorch and NumPy
- Conducted systematic experimental analysis across 5 dimensions (noise sensitivity, training set impact, parameter complexity, regularization effects) with comprehensive visualizations demonstrating theoretical ML concepts in practice

Variable Elimination for Bayesian Networks | Python, Probability and Statistics | GitHub 2024
- Implemented Variable Elimination Algorithm for Bayesian Networks to compute probability distributions with evidence
- Optimized sum-product computation for efficient inference in complex probabilistic networks

Dreamy Pastures (Multiplayer Board Game Engine) | Java, UML, OOP, Multiplayer | GitHub 2024
- Collaboratively developed local multiplayer board game supporting up to 4 concurrent players with real-time state synchronization and turn-based game logic, demonstrating OOP Design Patterns, and extensive UML Documentation

Design Patterns Library | Java, UML | GitHub 2024
- Created detailed library of OOP design patterns with UML diagrams and implementations, aligning with best practices.

RSA Encryption & Decryption System | Python, RSA, Security | GitHub 2021
- Implemented RSA cryptographic algorithm from mathematical foundations, including key generation using prime number factorization, modular exponentiation, and Euler's totient function with configurable prime digit lengths for customizable security levels
- Developed modular encryption system supporting both file and string encryption/decryption with Unicode numeral representation, integrating multiple in-house implementations of the algorithms (prime generation, modular arithmetic, extended GCD) into a cohesive cryptographic framework

Open Source Video Games | MBScript, Python, PHP | Floris Evolved | Tainted Paths | Medieval Conquests 2015-2019
- Founded and led development of three major game modifications with **314,000+ active subscribers** and **1.3+ million unique visitors** across platforms, achieving **90-95% positive ratings**
- Architected & developed approximately 100,000 lines of MBScript & Python code for game logic and modular systems
- Managed distributed teams of volunteers across multiple years, implementing complex features including custom gameplay mechanics and enhanced AI systems

Skills

Programming Languages: Python, Java, C, C++, BASH/Shell Scripting, MBScript, Assembly (MIPS, RISC-V), LaTeX, SQL
Frameworks & Libraries: PyTorch, TensorFlow, NumPy, Flask, Streamlit, YOLOv10, REST API, JDBC, PDFBox, Unified Modeling Language (UML)
Tools & Platforms: Git, PostgreSQL, Figma, Unity, Linux/Unix, Windows, AWS, Azure, Raspberry Pi, Arduino
Artificial Intelligence: Machine Learning, Deep Learning, Computer Vision (Neural Networks), Bayesian Networks, AI Agents (Information-Theoretic, Model-Based, Goal-Based, Utility-Based, Logic-based), Information Theory
Embedded Systems & Hardware: Hardware Design, Hardware Schematics, Debouncing, Automation
Software Engineering: Object-Oriented Programming (OOP), MVC Architecture, System Design, Parallel Processing, Procedural Programming, Logic Programming, Database Programming
Algorithmic Paradigms: Divide and Conquer, Greedy Algorithms, Dynamic Programming, Decrease and Conquer
Methodologies: Agile, Scrum, DevOps
Domain Expertise: Low-level programming, Game Engines, Security/Cryptography, Server Infrastructure, GIS, Digital Signal Processing, Computer Graphics, Haptics, Computer Networks, VR/AR, GPU, CPU
Languages: English (Native), Standard Arabic (Professional), Saudi Arabic (Professional), Darija (Native), French (MIFI Francisation, 2025-2026)

Relevant Coursework

Mathematical: Vector Calculus, Multivariable Calculus, Calculus 2, Applied Calculus, Advanced Linear Algebra, Linear Algebra, Mathematical Modeling, Probability and Statistics, Discrete Structures, Real Analysis, Mathematical Optimization, Modern Algebra, Differential Equations
Computer Science: Software Design, Operating Systems, Advanced Data Structures & Algorithms, Database Systems, Computer Networks, Graph Theory, Data Structures & Algorithms, Algorithm Analysis, Systems Programming, Digital Electronics and Digital Signal Processing, Artificial Intelligence, Simulation and Modeling, Computer Organization & Architecture, Object-Oriented Design, Introduction to Data Science, General Physics I, General Physics II.

Awards & Accomplishments

- **3rd Place Winner, The Pitch Competition ($1,000) –** Mount Allison University
  Secured recognition for presenting a scalable solution bridging AI and healthcare, showcasing hardware-software integration and algorithm design for medical assistance devices
- **Top 100 Global Ranking, League of Legends (EUW), Season 3 —** Ranked top 100 among 70M+ active players globally (top 0.000001%), demonstrating exceptional strategic decision-making and performance under high-stakes competitive pressure